Lutian Zhao

Contact Information

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Research Interests

Perverse sheaves, moduli of sheaves, and moduli of parahoric torsors; enumerative invariants of Calabi–Yau threefolds and surfaces.

Education and Positions

Kavli Institute for the Physics and Mathematics of the Universe, University of Tokyo

Project Researcher (August 2024 – present)

University of Maryland, College Park

Serguei Novikov Postdoctoral Fellow in Mathematics (September 2021 – August 2024)
Mentor: Amin Gholampour

University of Illinois, Urbana-Champaign

Ph.D. in Mathematics (August 2021)

- Dissertation: *Gopakumar-Vafa Invariants and Macdonald Formula*
- Advisor: Sheldon Katz

Irving Reiner Memorial Award in Algebra (2020)

Shanghai Jiao Tong University

B.A. in Mathematics, Zhiyuan College (May 2015)

- Highest honors in mathematics, highest distinction in general scholarship
- Kaiyuan Scholarship

Publications

Published and accepted

Huang, Pengfei, Georgios Kydonakis, Hao Sun, and Lutian Zhao. *Tame parahoric nonabelian Hodge correspondence on curves.* To appear in *Annali della Scuola Normale Superiore di Pisa, Classe di Scienze.* arXiv:2205.15475.

Kydonakis, Georgios, Hao Sun, and Lutian Zhao. *Logahoric Higgs torsors for a complex reductive group. Mathematische Annalen* **388** (2024), 3183–3228.

Kydonakis, Georgios, Hao Sun, and Lutian Zhao. *Poisson structures on moduli spaces of Higgs bundles over stacky curves. Advances in Geometry* **24** (2024), no. 2, 163–182.

Kydonakis, Georgios, Hao Sun, and Lutian Zhao. *Monodromy of rank 2 parabolic Hitchin systems. Journal of Geometry and Physics* **171** (2022), 104411.

Kydonakis, Georgios, Hao Sun, and Lutian Zhao. *The Beauville-Narasimhan-Ramanan correspondence for twisted Higgs V-bundles and components of parabolic* $\mathrm{Sp}(2n,\mathbb{R})$*-Higgs moduli spaces. Transactions of the American Mathematical Society* **374** (2021), no. 6, 4023–4057.

Kydonakis, Georgios, Hao Sun, and Lutian Zhao. *Topological invariants of parabolic G-Higgs bundles. Mathematische Zeitschrift* **297** (2021), 585–632.

Gong, Hongyu, Luoyi Fu, Xinzhe Fu, Lutian Zhao, Kainan Wang, and Xinbing Wang. *Distributed multicast tree construction in wireless sensor networks. IEEE Transactions on Information Theory* **63** (2017), no. 1, 280–296.

Preprints and work under review

Zhao, Lutian. *Topological string blowup equation via stable pairs.* arXiv:2608.25397 (2026).

Kydonakis, Georgios, and Lutian Zhao. *Level structures on parahoric torsors and complete integrability.* arXiv:2506.12302 (2025).

Zhao, Lutian. *Gopakumar-Vafa invariants and Macdonald formula.* arXiv:2306.05547 (2023).

Selected Talks

Vafa-Witten invariants, fixed loci, and ruled surfaces, Geometry, Algebra and Mathematical Physics Seminar, Nanjing University (May 2026)

Tame defects and parahoric Hitchin systems, Derived & Noncommutative Geometry Seminar, Shanghai Institute for Mathematics and Interdisciplinary Studies (February 2026)

Perverse sheaf Gopakumar-Vafa/Pandharipande-Thomas correspondence for local del Pezzo surfaces, Workshop on Mirror Symmetry and Related Topics, Kyoto University (December 2025)

Vafa-Witten invariants for ruled surfaces, GTM Seminar, Kavli IPMU (October 2025)

Perverse sheaf Gopakumar-Vafa/Pandharipande-Thomas correspondence for local del Pezzo surfaces, Counting Invariants and Enumerative Geometry, Brin Mathematics Research Center, University of Maryland (September 2025)

Moduli of parabolic Higgs bundles and Langlands duality, Tsukuba Workshop for Young Mathematicians, University of Tsukuba (February 2025)

Parahoric Higgs bundles and related geometry, GTM Seminar, Kavli IPMU (October 2024)

Geometry of parahoric Higgs bundles, Foundational Mathematics Research Seminar, Xi'an Jiaotong-Liverpool University (June 2024)

BPS counting from spectral networks, RIT on Geometry and Physics, University of Maryland (three talks, April–May 2024)

Nonabelian Hodge theory for parahoric torsors, Workshop on Geometric Representation Theory and Moduli Spaces, University of North Carolina at Chapel Hill (November 2023)

The journey through nonabelian Hodge theory: from vector bundles to parahoric torsors, Hua Luogeng Youth Forum, Academy of Mathematics and Systems Science, Chinese Academy of Sciences (May 2023)

Stability of parahoric Higgs bundles, Mid-South Algebraic Topology and Geometry Workshop (July 2022)

Gopakumar-Vafa invariants and support theorems, Algebra-Number Theory Seminar, University of Maryland (two talks, March–April 2022)

Supersymmetric Yang-Mills theory and geometric Langlands duality, RIT on Geometry and Physics, University of Maryland (February 2022)

Topological twists, RIT on Geometry and Physics, University of Maryland (February 2022)

Stability condition for tame parahoric Higgs bundles, Geometry and Topology Seminar, Chern Institute of Mathematics, Nankai University (November 2021)

Intersection cohomology and moduli space (mini-course), South China University of Technology (September 2021)

Gopakumar-Vafa invariants from Macdonald formula, Joint CUHK-Harvard-YMSC Differential Geometry Seminar, Harvard University (October 2020)

Gopakumar-Vafa invariants from Macdonald formula, Séminaire Arithmétique et géométrie algébrique, University of Strasbourg (February 2020)

Supersymmetry and Morse theory, Graduate Geometry and Topology Seminar, University of Illinois at Urbana-Champaign (October 2018)

Calculating Gopakumar-Vafa invariants, Summer School on Geometry of Moduli Spaces of Curves, ICTP, Trieste (June 2018)

Funding

JSPS KAKENHI Grant Number JP25K17226

Conferences Organized

The Geometry and Physics of Higgs Bundles, Kavli IPMU, 22–24 June 2026. With Hisashi Kasuya, Hiraku Nakajima, Laura Schaposnik and Mengxue Yang.

Seminars Organized

Sheldon Katz's Student Seminar (January 2017 – January 2021)

Graduate Student Algebraic Geometry Seminar (September 2017 – December 2018)

Geometric Langlands Learning Group (November 2016 – April 2017)

Courses Taught

University of Maryland, College Park

MATH 401, Applications of Linear Algebra (Spring 2022, Fall 2023)

MATH 406, Introduction to Number Theory (Fall 2022)

MATH 135, Discrete Mathematics for Life Sciences (Fall 2022, Spring 2023)

MATH 605, Homological Algebra (Fall 2023)

University of Illinois, Urbana-Champaign

MATH 234, Calculus for Business, discussion section (Spring 2017, Spring 2019)

MATH 220, Calculus I, discussion section (Fall 2017, Fall 2018, Fall 2019, Spring 2020)